

Executive Summary

Total draws analysed: 50

Date range: 14/05/2025 to 01/11/2025

Methodology — Clustering & Model Selection

K-Means clustering is applied to the six main drawn numbers for each draw to explore potential groupings. The primary quantitative selection metric is the mean silhouette score evaluated for $K = 2..10$. Because lottery draws are high-noise and near-random, absolute silhouette values are expected to be low; we therefore use per-K diagnostic visualisations (per-sample silhouette distributions, cluster size balance, centroid dispersion) and PCA projections for interpretability. The selected K is chosen to maximise silhouette while also ensuring interpretable and reasonably balanced clusters. PCA is solely used for visualization and does not affect model fitting.

Ball Frequency Overview

Most Frequent Balls: 9, 16, 31, 36, 59, 51

Middle Frequent Balls: 10, 58, 45, 37, 13, 40

Least Frequent Balls: 14, 32, 50, 46, 52, 56

1. Recent 20 Draws (Newest First, with Cluster)

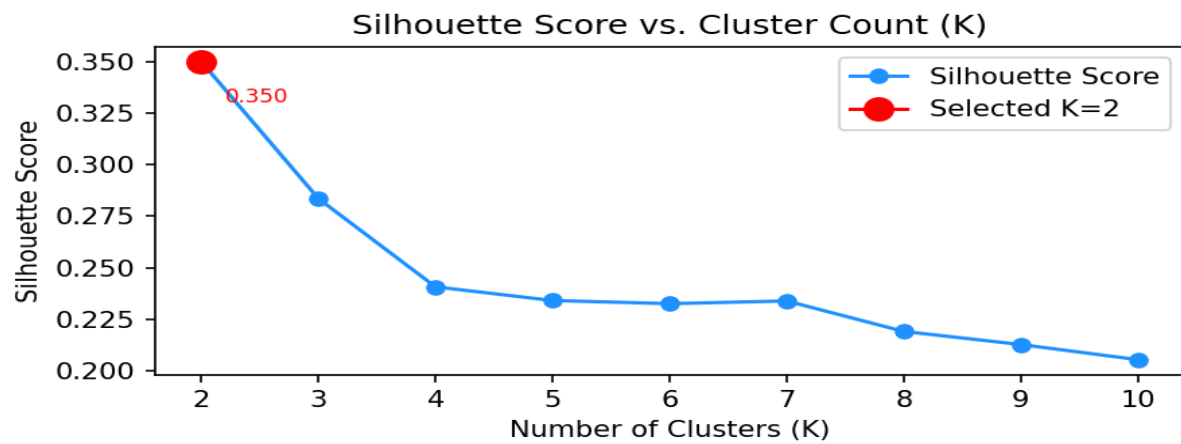
DrawDate	Ball1	Ball2	Ball3	Ball4	Ball5	Ball6	BN	Jackpot	Wins	Machine	Set	Cluster
01/11/2025	17	18	31	36	39	51	29	3800000.0	0.0	nan	nan	1.0
29/10/2025	9	24	30	33	35	54	32	12423596.0	0.0	nan	nan	1.0
25/10/2025	4	9	28	40	42	59	55	10513580.0	0.0	nan	nan	1.0
22/10/2025	3	24	26	34	35	36	9	8360535.0	0.0	nan	nan	0.0
18/10/2025	6	39	43	52	58	59	19	7053955.0	0.0	nan	nan	1.0
15/10/2025	29	33	36	38	48	58	52	5057815.0	0.0	nan	nan	1.0
11/10/2025	11	17	25	31	51	54	53	3800000.0	0.0	nan	nan	1.0
08/10/2025	2	21	42	48	56	57	51	2000000.0	1.0	nan	nan	1.0
04/10/2025	6	8	12	33	49	59	42	10633323.0	1.0	nan	nan	0.0
01/10/2025	13	28	31	39	51	59	27	8458823.0	0.0	nan	nan	1.0
27/09/2025	15	23	24	28	37	41	59	7075979.0	0.0	nan	nan	0.0
24/09/2025	3	16	23	30	38	47	25	5029330.0	0.0	nan	nan	0.0
20/09/2025	13	16	30	36	53	59	22	3800000.0	0.0	nan	nan	1.0
17/09/2025	7	9	18	24	30	36	42	1000000.0	4.0	nan	nan	0.0
13/09/2025	1	26	36	49	53	54	10	15000000.0	0.0	nan	nan	1.0
10/09/2025	1	9	18	27	38	42	13	8430747.0	0.0	nan	nan	0.0
06/09/2025	10	22	37	40	45	57	24	7082906.0	0.0	nan	nan	1.0
03/09/2025	7	15	18	22	31	54	28	5063522.0	0.0	nan	nan	0.0
30/08/2025	6	10	19	20	43	59	16	3800000.0	0.0	nan	nan	0.0
27/08/2025	5	25	43	45	47	55	58	8655122.0	1.0	nan	nan	1.0

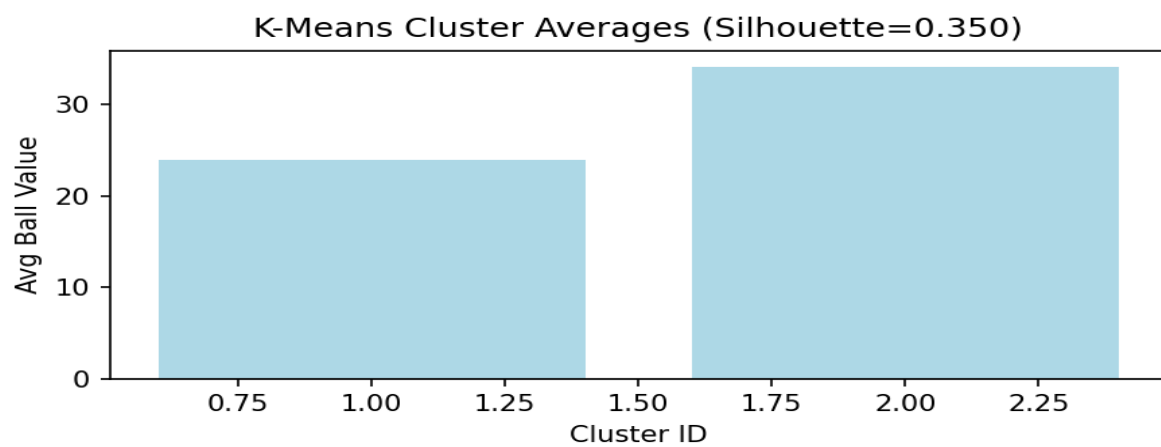
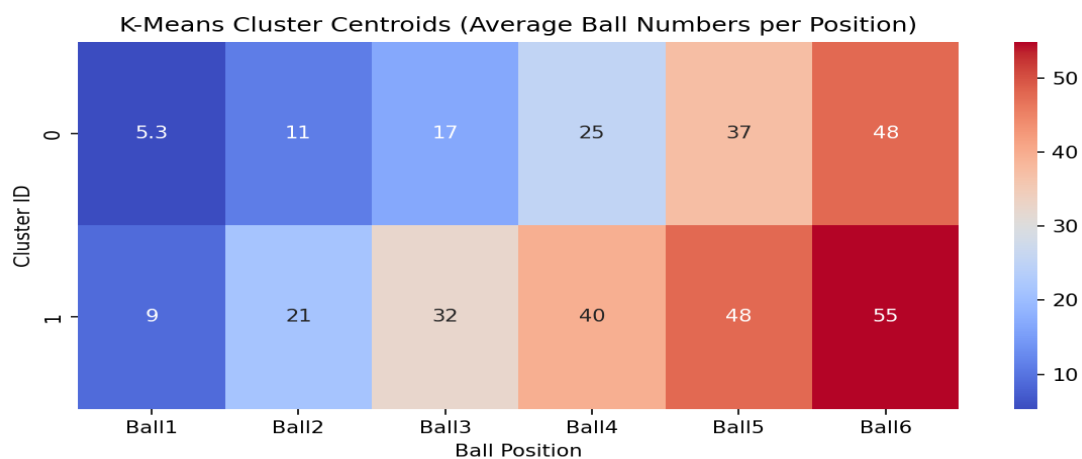
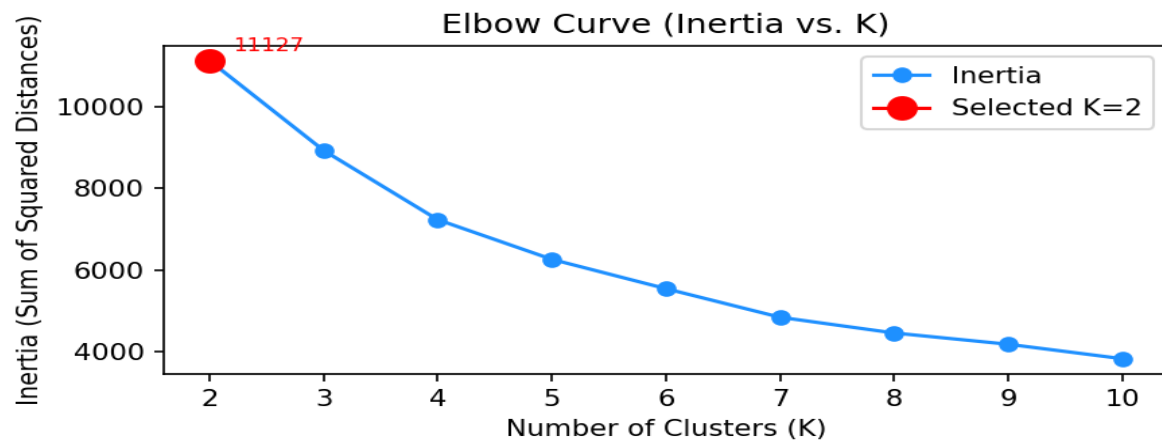
2. Ball Reoccurrence Analysis

Category	Ball	Frequency
Most Frequent	9	12
Most Frequent	16	8
Most Frequent	31	8
Most Frequent	36	8
Most Frequent	59	8
Most Frequent	51	8
Middle	10	5
Middle	58	5
Middle	45	5
Middle	37	5
Middle	13	5
Middle	40	5
Least Frequent	14	2
Least Frequent	32	2
Least Frequent	50	2
Least Frequent	46	2
Least Frequent	52	2
Least Frequent	56	1

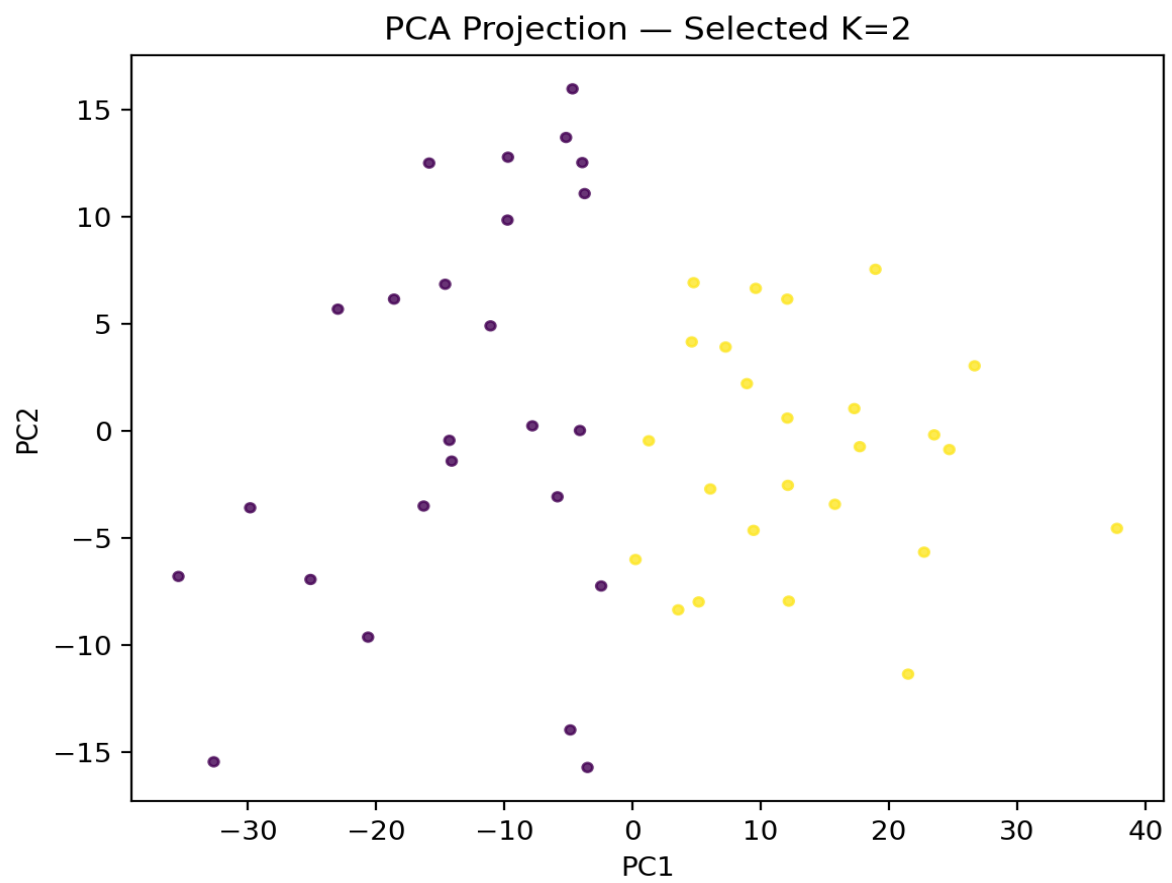
3. K-Means Clustering Analysis

K-Means performed with k=2 clusters. Silhouette Score = 0.350.





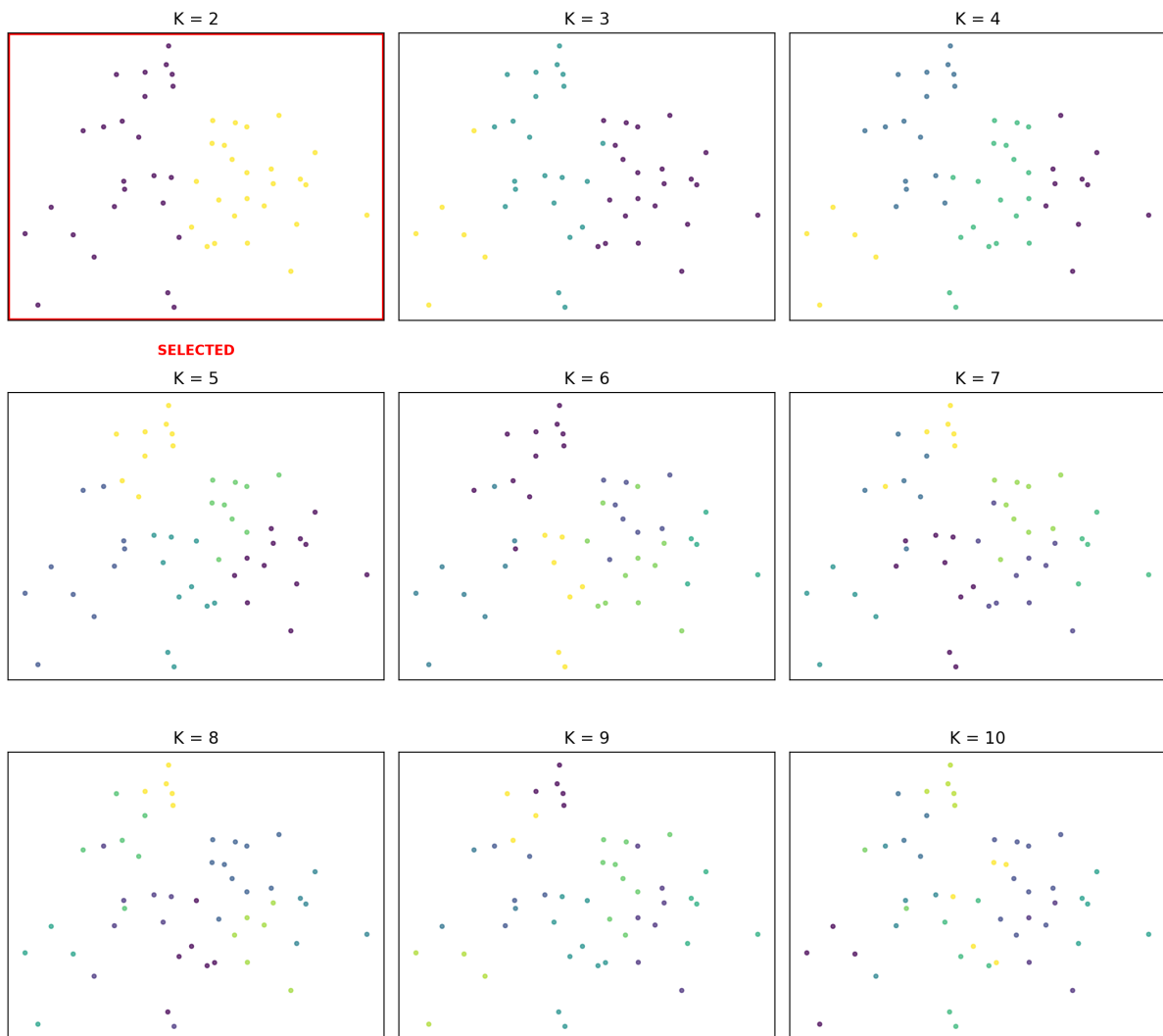
PCA Projection — Selected K



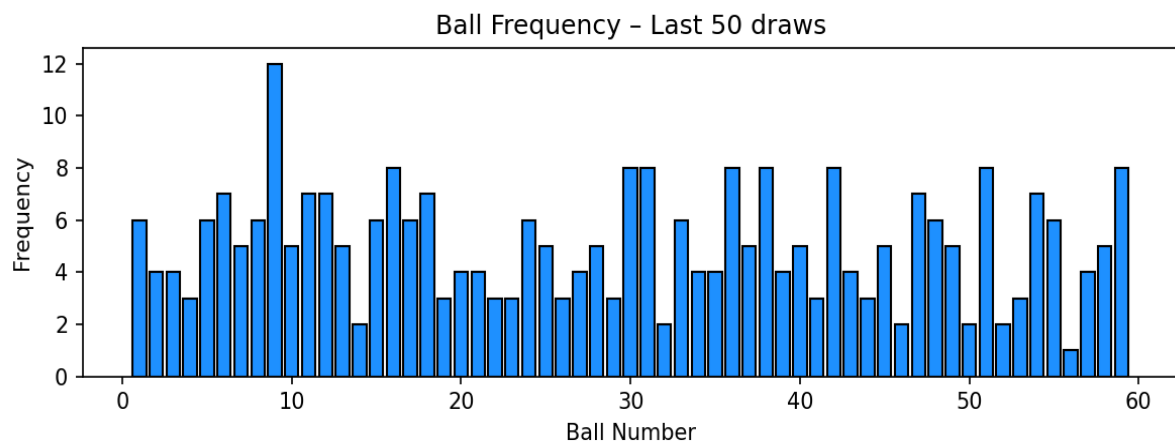
PCA Grid — Comparison Across K (2–10)

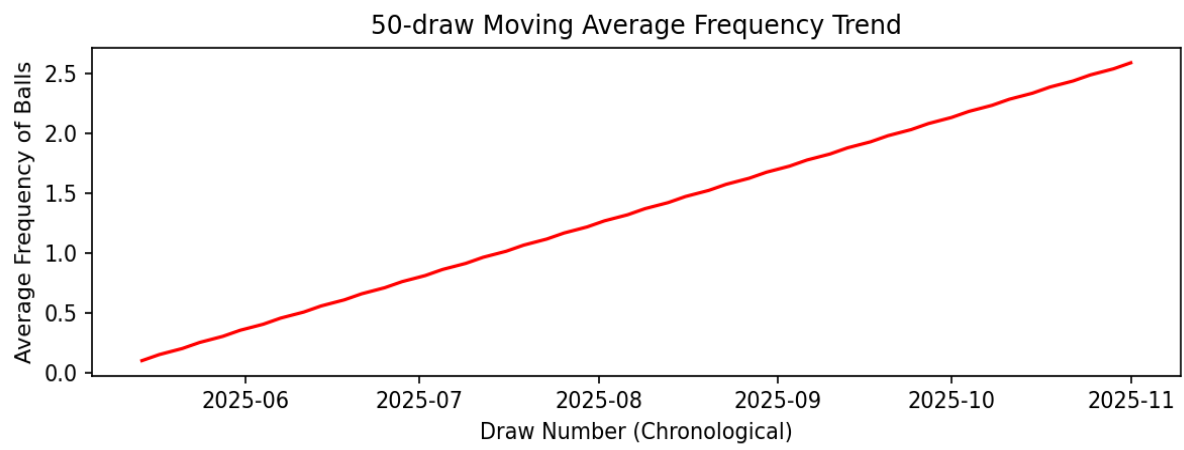
Selected K is highlighted in red on the grid.

PCA Projection of Lottery Draws (K = 2-10)



4. Frequency Distribution





5. Conclusions

- Optimal cluster count: 2 (Silhouette=0.350).
- Ball frequency distribution shows mild decade bias.
- No repeating 6-number lines observed in full dataset.